Experiment Protocol: Validating TACS and IPR (Multimodal Entity Indexing)

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Disclosure: The author founded and leads PahamSEO, an Indonesian digital education community (since 2017) that conducts independent research and offers free training and coaching for MSMEs. The findings may inform the community's educational materials, so a non-financial interest exists. To limit its influence, the research is hosted on a separate domain, and hypotheses and the analysis plan are registered before data collection.

Purpose and scope

This experiment tests one question: does a coordinated set of three text anchors (TACS) make an image easier to find and cite than conventional practice? It is deliberately small, 300 images, so a single researcher can run it cleanly and others can replicate it.

Research questions:

1. Does full TACS raise the chance that an image appears in image search results for relevant queries?
2. Which anchor (file name, alt text, caption) contributes to the effect, alone or together?
3. Does the reformulated IPR score predict the outcome?

Out of scope: video, audio, and general claims about all generative engines. All text, queries, and searches are in English, so no claim is made for other languages. Every result, including a null or negative one, is reported.

Related work

This study sits next to several recent lines of work. The summaries follow the authors' own abstracts; none of these studies tests the question asked here.

- **Generative Engine Optimization.** Aggarwal et al. (2023) introduced GEO and reported that content-level changes to a source can raise its visibility in generative engine responses [1]. This concerns page text in generated answers, not image metadata.
- **Joint image-text embeddings.** Radford et al. (2021) trained CLIP, a joint embedding of images and text that underlies much of modern text-to-image retrieval [2].
- **Caption Injection.** Chen and Liao (2025) propose a multimodal GEO method that extracts captions from images with a vision-language model and works them into the page text, evaluated on a benchmark of generated answers [3]. Here the image's own metadata anchors are manipulated and the outcome is appearance in live image search, so the manipulation and the outcome both differ.
- **Multimodal GEO.** Du et al. (2026) alternately optimize image perturbations and text suffixes to shift the rankings of vision-language model rankers over product items [4]. That is adversarial optimization against a model ranker; this study tests ordinary metadata practice in a live search engine.
- **Critical survey.** Martinez (2026) reviews 45 studies and argues that the gains reported for the foundational GEO method apply when a source is already in a fixed context, not to organic discoverability; it reports no technique with a stable, cross-platform causal effect on organic discoverability and proposes an evidence hierarchy and a protocol that fixes the estimand, design, and analysis plan before data collection [5]. This protocol follows that spirit.

What this study adds, stated narrowly. To the author's knowledge, no controlled field experiment has tested file name, alt text, and caption, alone and combined, on whether an image appears in live image search. This statement is limited to the work the author could find and may be wrong; readers are asked to point out missing work. The design is close to a randomized field experiment but has no platform logs, so it does not claim the highest evidence level described in [5].

Hypotheses

The three hypotheses below are written before any data are collected, each with the result that would reject it. Minimum effect sizes must be confirmed in the pilot (see Sample and data).

ID	Statement	Result that rejects it	Minimum meaningful effect
H1	Images with full TACS (C4) appear in the top 10 of image search more often than conventional controls (C0).	Difference in proportions is not distinguishable from 0, or the 95% confidence interval includes 0.	15 percentage points
H2	File name, alt text, and caption each raise the chance of appearing relative to C0.	No single anchor differs from C0 after Holm correction.	10 percentage points per anchor
H3	The reformulated IPR score predicts the chance of appearing, controlling for condition and category.	The IPR coefficient is not distinguishable from 0 on held-out data not used to choose the formula.	Odds ratio 1.3 per 1 SD increase in IPR

H1 is the only primary confirmatory hypothesis. H2 and H3 are secondary. Interaction between anchors (is C4 larger than the sum of the single-anchor effects) is reported as exploratory.

Design

The design is a controlled experiment with five conditions in which only the three text anchors differ and everything else is identical. Each image is assigned one condition at random.

Condition	File name	Alt text	Caption
C0 Conventional control	Short descriptive (for example `sales-chart.webp`)	One short phrase	None
C1 File name only	TACS format	As C0	None
C2 Alt text only	As C0	TACS format (deep alt text)	None
C3 Caption only	As C0	As C0	TACS format
C4 Full TACS	TACS format	TACS format	TACS format

Rules that protect validity:

- **One image, one page, one condition.** Each image sits on its own page with the same template, text length, and heading structure. Surrounding text is identical across conditions.
- **Stratified randomization.** Randomize within image category (for example diagram, product photo, infographic) so every condition has the same composition. Record the random seed.
- **Simultaneous publication.** All pages go live on the same day in random order, so changes in an algorithm or model do not fall on one condition.
- **Blind annotation.** The person who writes the TACS text and the person who writes queries or codes outcomes are different people, and coders do not know the condition.
- **Language lock.** All file names, alt text, captions, page text, and queries are in English. Searches use an English interface and one fixed region (recorded). Any claim is limited to English-language search.

Sample and data

The planned sample is 300 original images, 60 per condition. With that size, an effect of about 22 percentage points from a 15% baseline is detectable at 80% power. This is a rough two-proportion estimate without corrections, so verify it with G*Power or statsmodels before locking the sample size.

- **Image source.** Self-made images (photos, diagrams, infographics) never published or uploaded anywhere, so no existing index entry or duplicate exists. Split evenly into three categories of 100 images.
- **Rights and ethics.** No faces, personal data, or third-party brands. Page content must be real and not misleading, not spam.
- **Separate hosting.** Use the dedicated research domain, one domain for the whole sample, so domain authority is the same for every condition.
- **Pilot first.** Run 30 images (6 per condition) for two weeks to estimate the baseline appearance rate, test the measurement procedure, and confirm the outcome is not zero in every condition. If the baseline is near 0%, change the outcome definition or queries before the main run, and update the pre-registration.
- **Stored raw data.** Original image files, the text for each condition, publication dates, the randomization list with seed, and every measurement per image per time point.

Outcome measures

The primary outcome is whether an image appears in the top 10 of image search for a relevant query, because it is observable, binary, and not directly affected by the treatment. Citation in generative answers is too rare and too hard to observe to be primary, so it is secondary.

Role	Metric	How measured	Note
Primary	Appears in image search top 10	Binary per image: appears at least once for any of its queries on day 28	The only confirmatory outcome
Secondary	Image indexed	Image index status from the search engine's official tool	Stage before appearing; reported separately
Secondary	Offline retrieval recall@k	An evaluator embedding model retrieves images using blind annotators' queries	Evaluator model differs from the model used to compute IPR
Secondary	Citation in generative answers	Image or page appears as a source in an AI answer for the same queries	Rare event; report as observed

Two hard rules prevent circularity:

- The embedding model used to compute IPR and the evaluator model used in offline retrieval must differ (for example OpenCLIP for IPR and SigLIP for the evaluator).
- Alt text must not be optimized against the evaluator model. A higher cosine similarity on the same model used to write the text only shows the text matches that model, not that the image is more visible in a search engine.

Measurement protocol

Search results vary by time, location, and personalization, so every measurement is repeated and logged so that variation can be separated from the treatment effect. Measurement follows these steps.

1. **Blind query set.** Two annotators who see only the image, not its file name, alt text, or caption, each write three natural English queries per image. Merge and deduplicate. Queries must not be written by the person who wrote the TACS text.

2. **Schedule.** Measure on days 7, 14, 28, and 56 after publication. Day 28 is the primary time point; the others show dynamics.
3. **Repeats.** Run each query five times on different days and hours within the measurement window, and record the share of repeats in which the image appears.
4. **Uniform conditions.** Clean browser profile, no login, fixed location, English interface and region. Record device and language settings. If possible, add a second location as a robustness check.
5. **Raw evidence.** Save screenshots or result HTML, date and time, and the engine or model version when known.
6. **Blind coding.** The person who codes the outcome (appears or not) does not know the image's condition. Code from evidence files whose names have been randomized.
7. **Compliance.** Use official APIs or tools where available, and check each engine's terms of service before automating. Manual measurement is acceptable if the query volume is manageable.

IPR reformulation

IPR must be reformulated before it is tested. The current version adds an OCR entity density and a cosine similarity score, two quantities with different scales and units, and K_noise and D_pixel have no measurable definition. The steps below turn it into a metric that can be validated.

1. **Operational definitions.** E_ocr is the number of verified factual entities (numbers, names, units), not raw OCR character count. D_pixel is file size and megapixels, normalized. K_noise is the share of image area judged uninformative by three annotators; report inter-annotator agreement (for example ICC).
2. **Normalization.** Convert each component to a 0 to 1 scale against a reference corpus (percentile rank, not raw min-max) so components can be combined.
3. **Two candidate forms.** Write both forms below and choose between them using a criterion fixed before data: which better predicts the outcome on training data, then evaluate on separate test data.
4. **Construct validation.** Correlate IPR with human ratings of the image's information density. If the correlation is low, IPR does not yet measure what it claims.
5. **Data split.** Use 70% of images to choose the formula's form and 30% to test H3. Never use the test data to select the form.

```
\text{IPR}_A = \frac{w_1 \hat{E}_{ocr} + w_2 \hat{S}_{clip}}
{\hat{D}_{pixel}\,(1 - \hat{K}_{noise}) + \varepsilon}
```

```
\text{IPR}_B = \frac{\hat{E}_{ocr} \cdot \hat{S}_{clip}\,(1 -
\hat{K}_{noise})}{\hat{D}_{pixel} + \varepsilon}
```

A hat marks a component normalized to 0 to 1, and epsilon is a small constant so the denominator is never zero. The weights w1 and w2 must be fixed or estimated from training data only.

Analysis plan

All analyses are written and registered before the first data point is collected, and only one comparison is confirmatory. This prevents choosing a test or an outcome after seeing the data.

Analysis	Method	Reporting
Primary (H1): C4 vs C0, in top 10 on day 28	Logistic regression with condition, image category, and publication batch as covariates	Difference in proportions and odds ratio with 95% CI; Fisher exact test as a check
Secondary (H2): C1, C2, C3 vs C0	Same model with Holm correction for three comparisons	Effect sizes and intervals, adjusted p-values
Secondary (H3): IPR as predictor	Logistic regression on the 30% test data, controlling for condition and category	Odds ratio per 1 SD of IPR with interval
Repeated measures	Mixed model (GLMM) with a random intercept per image	Share of repeats in which the image appears
Exploratory	Anchor interactions, results by category, other time points	Labelled exploratory, no confirmatory claims

Decision rules fixed in advance:

- **H1 supported** if the difference is at least 15 percentage points and the lower bound of the 95% interval is above 0.
- **H1 not supported** if the interval includes 0. Report as a null result, not as "insufficient data".
- **One analysis point.** The confirmatory analysis is run once, on day 28 data, with no interim peeking.
- **Images never indexed** count as not appearing in the primary outcome and are reported separately under the secondary indexing outcome.

- **Exclusion rules** (for example blocked or broken pages) are written before data collection, and every exclusion is reported with its reason.

Threats to validity

The largest threats are metric circularity and confounding from content and domain, and both are prevented by design rather than by later analysis. The table maps each threat to its mitigation.

Threat	Cause	Mitigation
Circularity	The same embedding score is used to write text and to measure outcome	Different models for IPR and the evaluator; text not optimized against the evaluator
Content differs across conditions	Images or surrounding text are not equivalent	Stratified randomization, identical page template and surrounding text
Domain authority	An established domain (such as pahamseo.org) influences ranking	A new research domain, the same for all conditions
Drift over time	Algorithm or model changes during the experiment	Simultaneous publication; all conditions measured on the same days
Query bias	Queries written by someone who knows the TACS text	Blind annotators who see only the image
Rater bias	Coders know the image's condition	Evidence files with randomized names, blind coding
Engine variability	Results differ by time, location, personalization	Five repeats, uniform conditions, location recorded
Baseline near zero	Almost no images appear in any condition	30-image pilot; change the outcome definition before the main run if needed
Limited generalization	One language, one domain, three image categories	State as limitations; do not claim all engines, languages, or domains
Multiple comparisons	Many tests, many time points	One confirmatory test; Holm correction for secondary; rest exploratory

Reproducibility and publication

The strongest evidence an independent researcher can offer is a protocol others can inspect before results exist. Pre-registration and external review therefore come before data collection.

1. **Pre-registration.** Register hypotheses, design, metrics, and the analysis plan on OSF Registries (osf.io) before publishing the test pages. The timestamp shows the analysis was not changed after seeing data.
2. **External protocol review.** Ask two researchers with information retrieval or statistics backgrounds to read this document before data collection, and record their comments and the changes made. Review requests were sent on 29 October 2026; comments and the resulting changes will be listed in the version history.
3. **Open repository.** Code, the query set, the per-image condition list with seed, raw measurement data, and screenshot evidence go to GitHub, then are released on Zenodo to obtain a DOI. The protocol itself is archived the same way (v0.1: 10.5281/zenodo.23757281; source: github.com/multimodalentity/protocol).
4. **Preprint.** Upload the report to arXiv (category cs.IR; endorsement may be needed) or OSF Preprints, linking the pre-registration and the data DOI.
5. **Full reporting.** Report all conditions, all time points, and all deviations from the protocol, including null or negative results.
6. **Correct existing claims.** Until results exist, the empirical figures previously shown on the older dissertation page must stay removed or be labelled as unsupported by published data. Keep the web page aligned with the conceptual PDF.
7. **Versioning.** Publish version 1 as a conceptual monograph and version 2 as the empirical report, each with its own DOI.

Schedule and checklist

From protocol to preprint the plan takes about 18 weeks, and the order must not be reversed: the protocol and pre-registration are finished before the test pages are published.

1. **Weeks 1 to 2: Finalize the protocol.** Lock hypotheses and sample size, request external review, register on OSF.
2. **Weeks 3 to 4: IPR and pilot.** Reformulate IPR, validate with annotators, run the 30-image pilot.
3. **Weeks 5 to 6: Production and publication.** Create 300 images, write the text for each condition, randomize, and publish all pages on the same day.
4. **Weeks 7 to 14: Measurement.** Measure on days 7, 14, 28, and 56 as specified.
5. **Weeks 15 to 16: Analysis.** Run the registered analyses with no added confirmatory tests.
6. **Weeks 17 to 18: Report.** Write the report, release data to Zenodo, upload the preprint.

Checklist before publishing the test pages:

- [ ] Hypotheses, metrics, and analysis plan are registered on OSF
- [ ] Two external reviewers have read the protocol
- [ ] The separate research domain is live
- [ ] The evaluator model differs from the model used for IPR
- [ ] The pilot shows the baseline is not near zero
- [ ] Query annotators are blind to the TACS text
- [ ] All pages use the same template and identical surrounding text
- [ ] All text, queries, and searches are in English
- [ ] Publication order is randomized and all pages go live on the same day

References

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2. Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., ... & Sutskever, I. (2021). Learning transferable visual models from natural language supervision. Proceedings of ICML, PMLR 139, 8748-8763. https://doi.org/10.48550/arXiv.2103.00020
3. Chen, X., & Liao, Y. (2025). Caption Injection for Optimization in Generative Search Engine. arXiv:2511.04080. https://doi.org/10.48550/arXiv.2511.04080
4. Du, Y., Yu, C., Xu, H., Wang, Z., Zhao, Y., & Hu, X. (2026). Multimodal Generative Engine Optimization: Rank Manipulation for Vision-Language Model Rankers. arXiv:2601.12263. https://doi.org/10.48550/arXiv.2601.12263
5. Martinez, O. (2026). Optimizing Visibility in Generative Engines: A Critical Survey of Generative Engine Optimization (2023-2026). arXiv:2607.14035. https://doi.org/10.48550/arXiv.2607.14035

Changes from v0.1

- Added Related work and References.
- Softened the novelty wording and stated the evidence level the design can claim.
- Updated the header (version, date, review status) and added archive DOIs to Reproducibility and publication.
- No change to hypotheses, design, outcome measures, sample size, or analysis plan.